

# Measurement-Efficient Port-Reading Policies Policies for Fluid Antenna Systems Under Spatially Correlated $\kappa$ – $\mu$ Shadowed Fading

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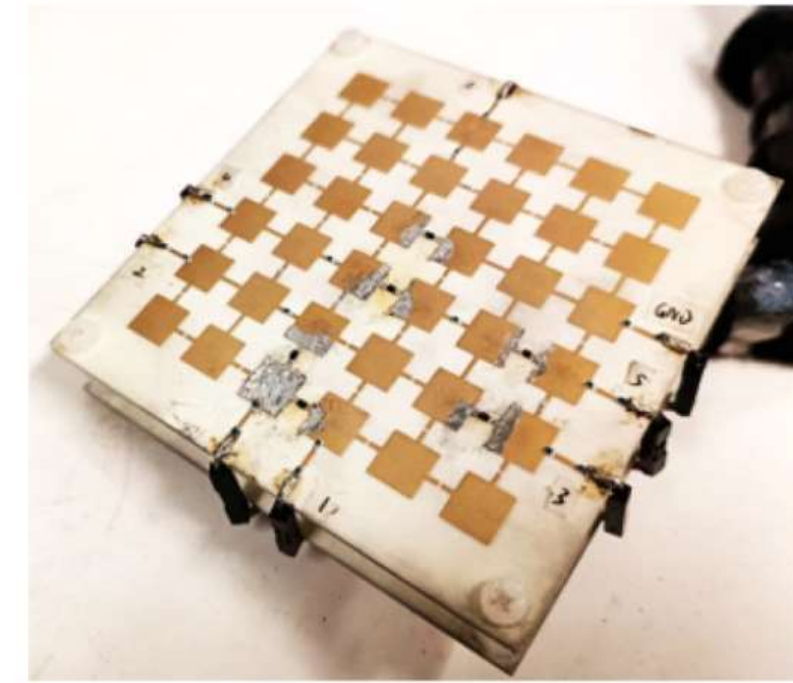
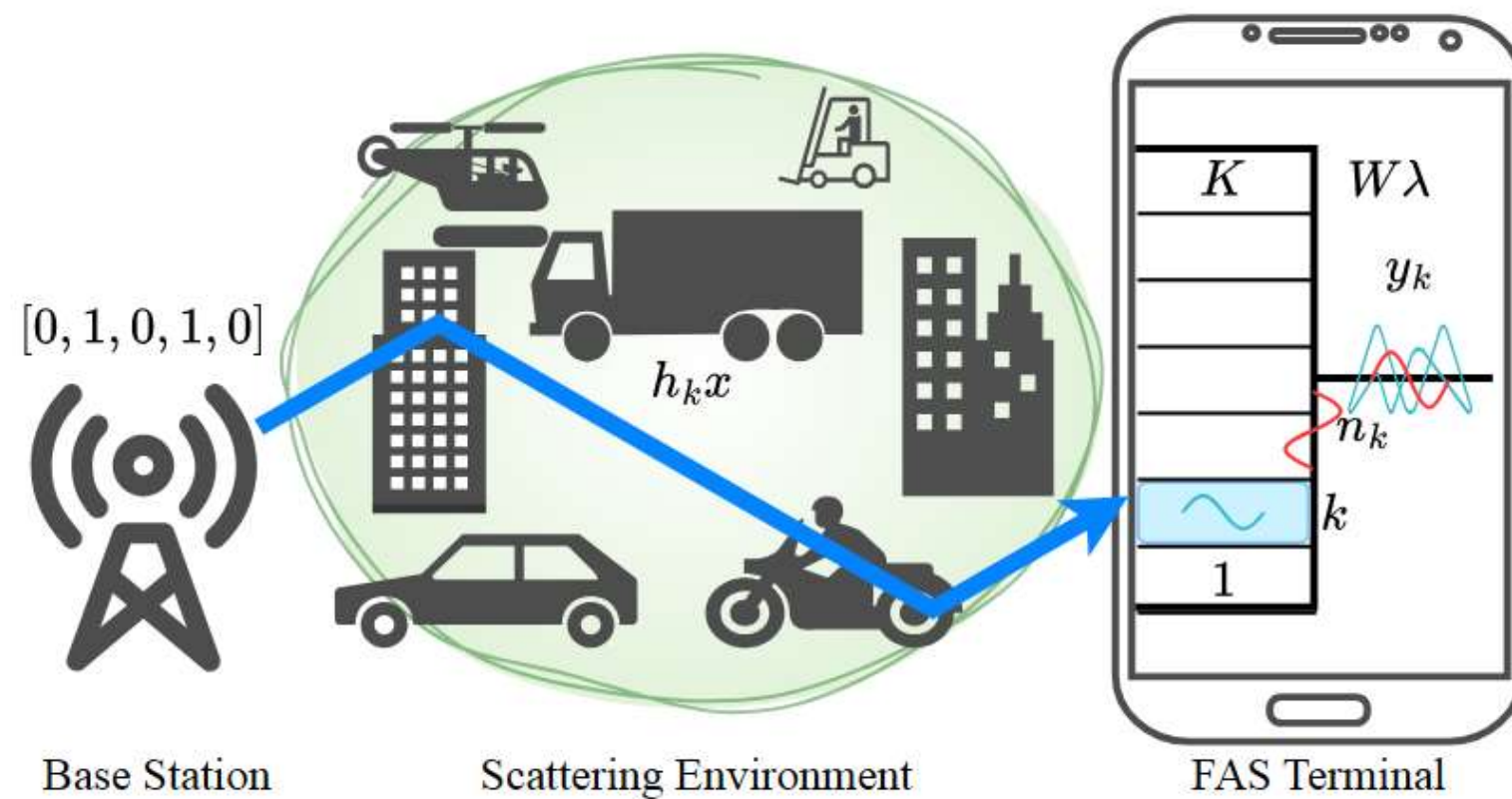
## Presentation Outline

- Fundamentals: FAS, Shadowed Fading, & the Port-Reading Problem
- ML-Based Channel Prediction from Sparse Measurements
- Synergy: How Optimized Reading Patterns Enhance ML Predictor Accuracy and Efficiency
- Core Contribution & Evaluation

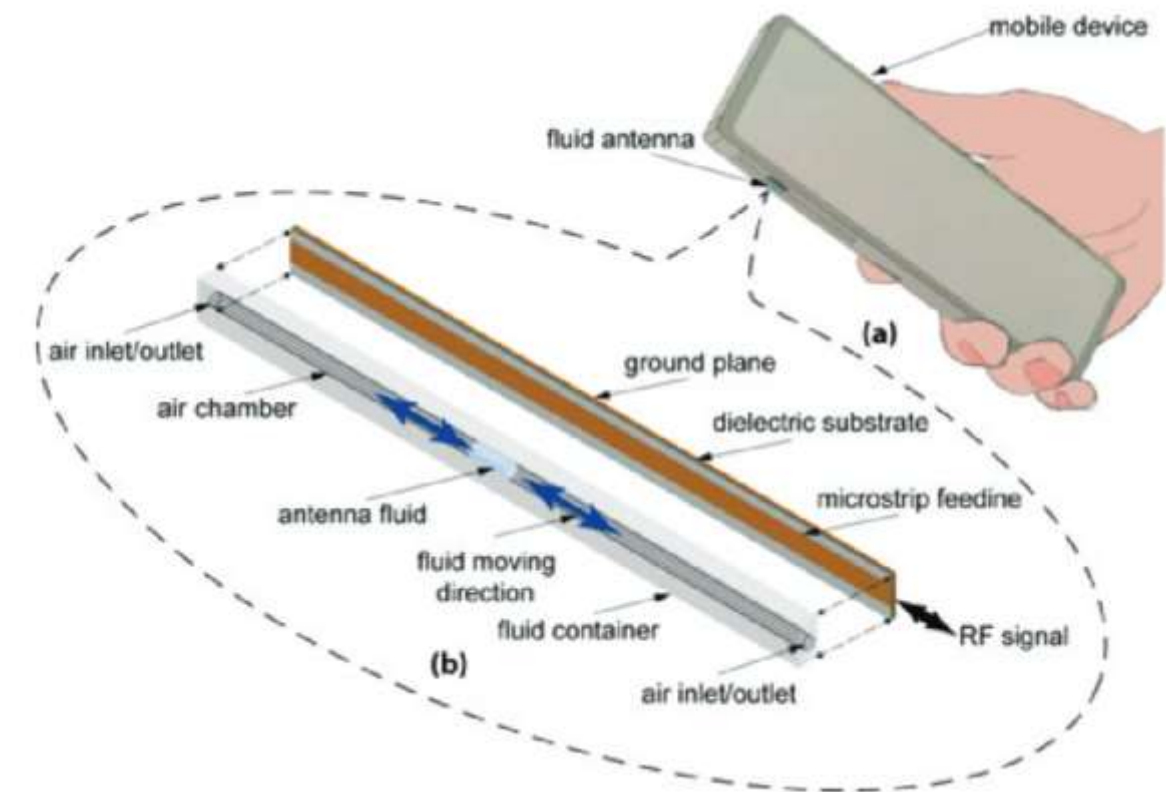


# Fluid Antenna System

- The concept of the **Fluid Antenna System (FAS)** emerged to improve performance.
- (2020) Kai Kit Wong



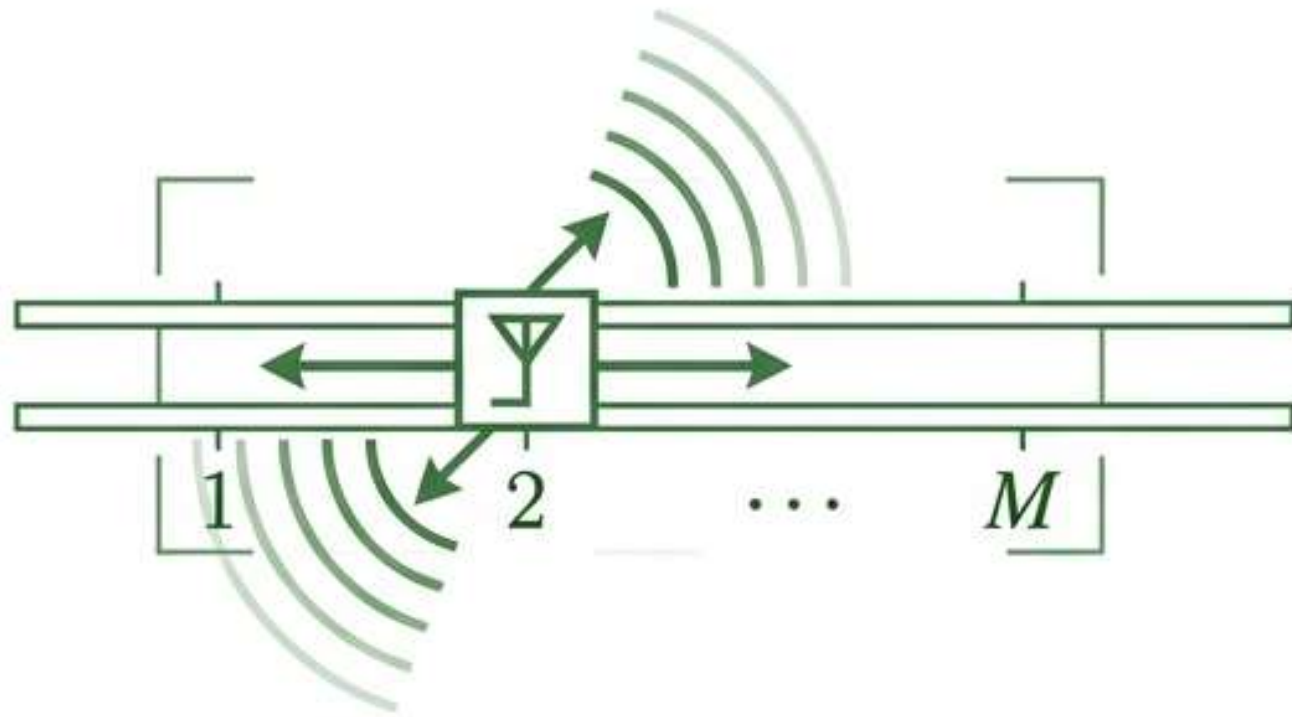
Antenna relocation can be achieved by mechanically moving liquid conductors through a microfluidic system, or by electronically switching radiating pixels on and off.





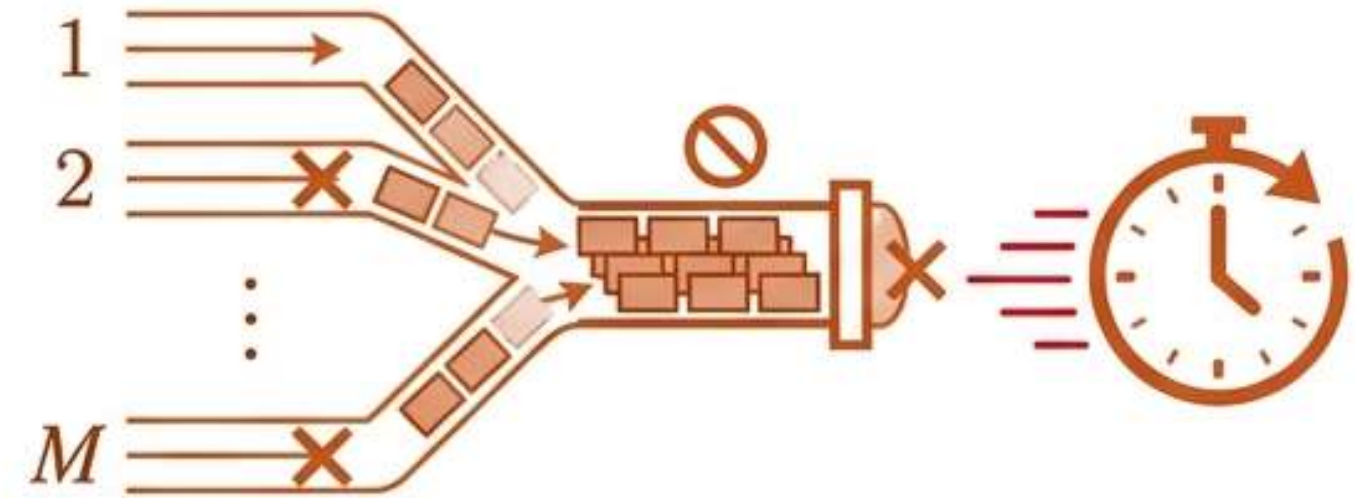
# Fluid Antenna Systems (FAS): Massive Diversity vs. Massive Overhead

## The Promise



A single RF chain electronically moves among  $M$  candidate positions within a small space. Approaches or exceeds classical multi-antenna combining gains without the massive hardware footprint.

## The Bottleneck

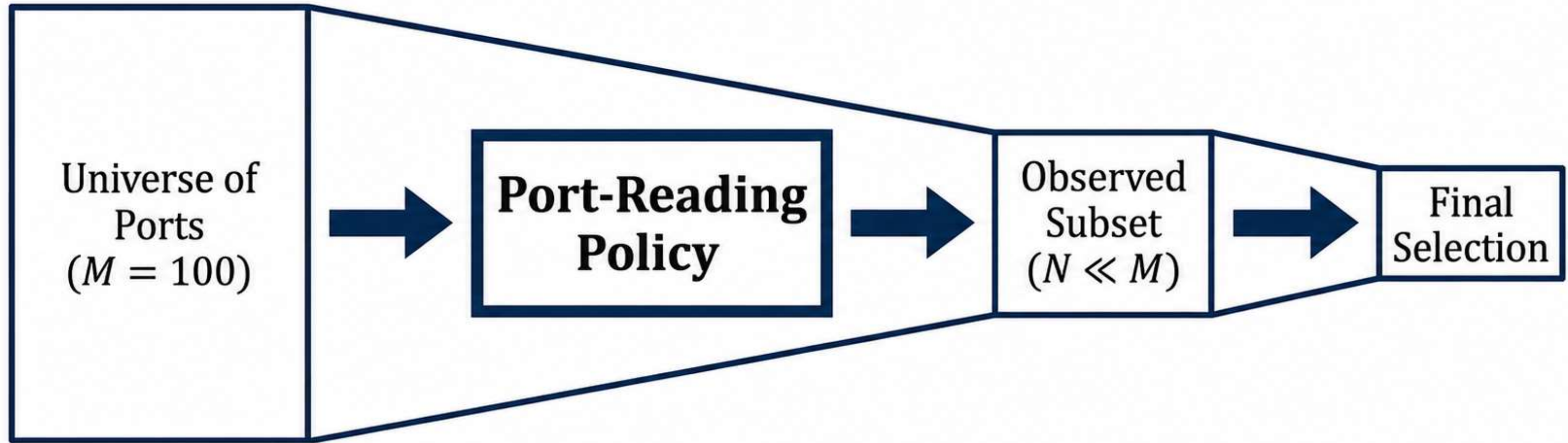


Reading all  $M$  ports induces severe measurement overhead, switching delay, and heavy receiver complexity.

**THE RESEARCH GAP:** Current machine learning approaches focus on selecting ports after all measurements are taken. But what if strict latency constraints mean we can only afford to measure a tiny fraction?



# Isolating the Sensing Pattern from Downstream Selection



## Core Objective

Treat port reading as a distinct policy-level design problem for measurement-limited FAS.

## Key Question

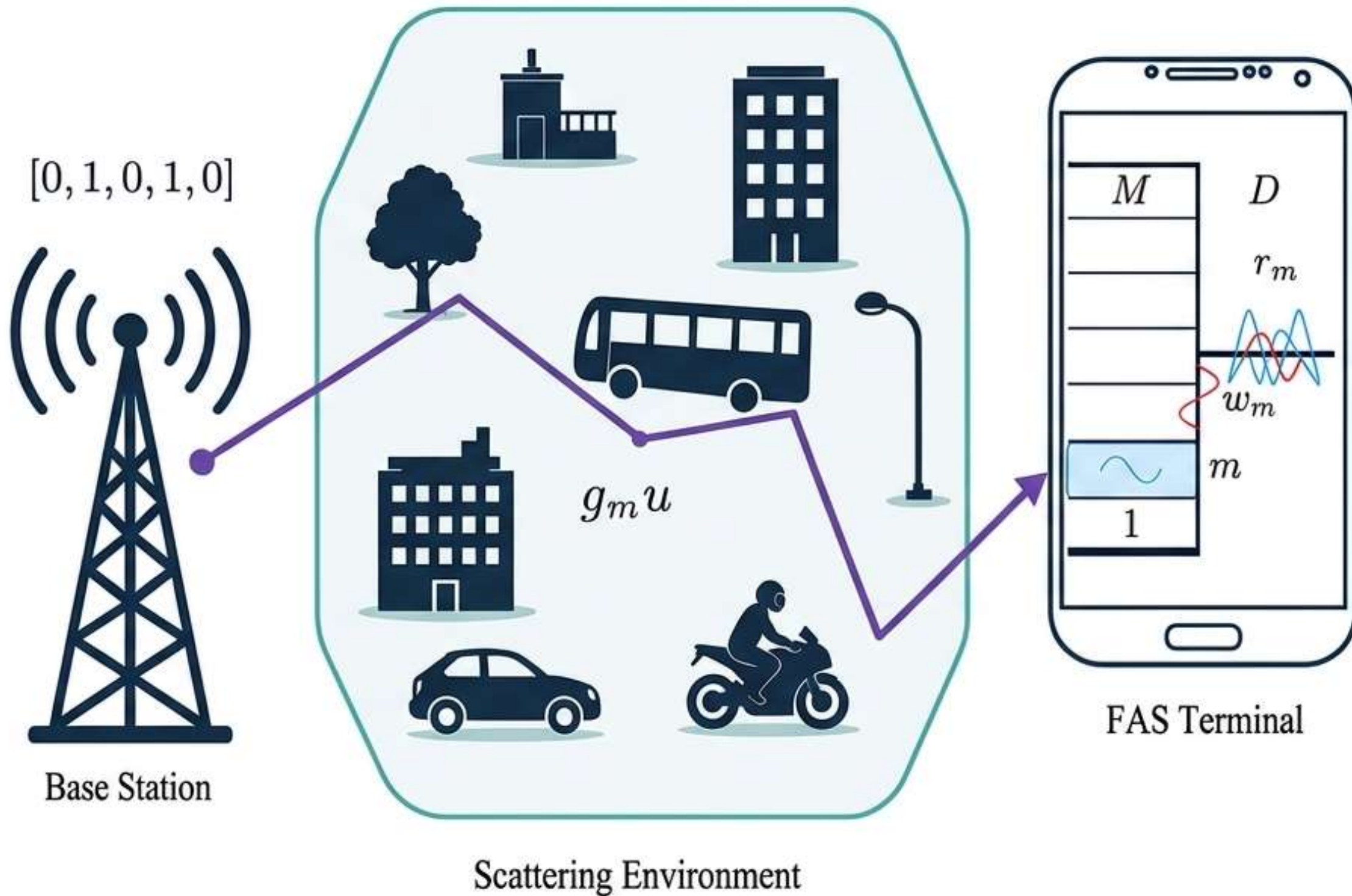
Should the measured set prioritize geometric coverage, marginal port quality, or complementarity?

## Evaluation Metric

Observed-only outage probability. The receiver selects strictly from the measured ports, isolating the efficacy of the reading policy from downstream reconstruction algorithms.



# System Model: Single-Antenna BS to a Multi-Port FAS Terminal



## Mathematical Constraints

$M$ : Candidate ports over normalized aperture  $D = L/\lambda$

$r_m = g_m u + w_m$ : Baseband sample at port  $m$

$\eta$ : Full Signal-to-Noise Ratio (SNR) profile

$\mathcal{O}$ : Index set of measured ports ( $N \ll M$ )

The receiver operates strictly from observation vector  $\eta_{\mathcal{O}}$ , never the full profile  $\eta$ .



# Spatially Correlated $\kappa$ - $\mu$ Shadowed Fading Environment

## Correlation & Power Models

Sub-wavelength spacing forces high correlation, modeled via a Jakes-type matrix:

$$[R]_{m,n} = J_0(2\pi|m - n|/D(M - 1)).$$

Power Model:

$$|g_m|^2 = \sum_{l=1}^{\mu} |q_{m,l} + p_m|^2.$$

|           |                      |                                                          |
|-----------|----------------------|----------------------------------------------------------|
| $p_m$     | Dominant Component   | Perfectly correlated, shifts entire aperture coherently. |
| $q_{m,l}$ | Scattered Components | The primary source of spatial diversity.                 |

## Dataset Regimes

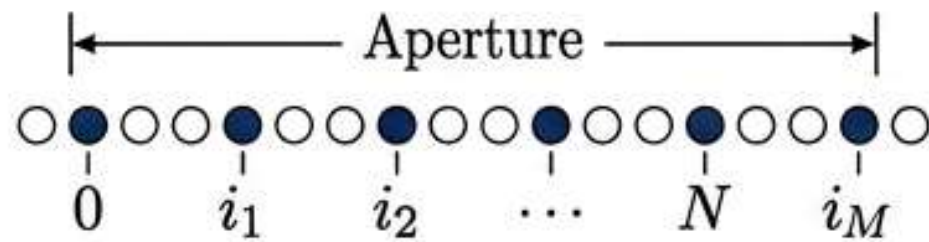
Generated 14 diverse fading regimes manipulating dominant-to-scattered power ( $\kappa$ ), clusters ( $\mu$ ), and shadowing severity ( $m$ ).



Ensures evaluation robustness across heterogeneous real-world conditions.

# Three Approaches to the Port-Reading Problem

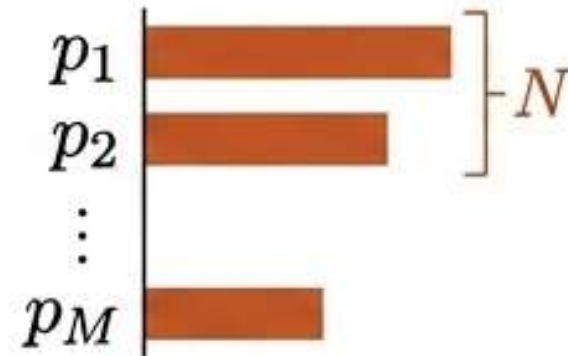
## Policy A (Uniform Reading)



**Method:** Subsamples the aperture by evenly spacing  $N$  indices.

**Focus:** Geometric spatial coverage. Deterministic and requires no data-driven tuning.

## Policy B (Marginal Ranking)

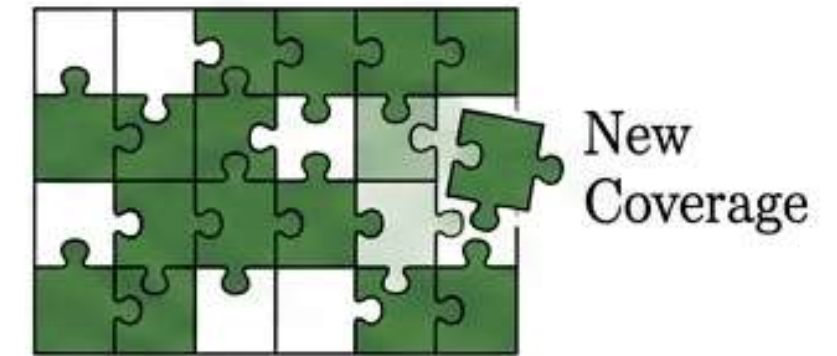


**Method:** Scores and ranks ports based on historical exceedance:

$$\hat{p}_m = \frac{1}{|\mathcal{D}_{des}|} \sum b_{i,m}$$

**Focus:** Threshold-aware per-port individual quality.

## Policy C (Greedy Coverage)



**Method:** Iteratively adds ports that maximize the number of newly covered samples.

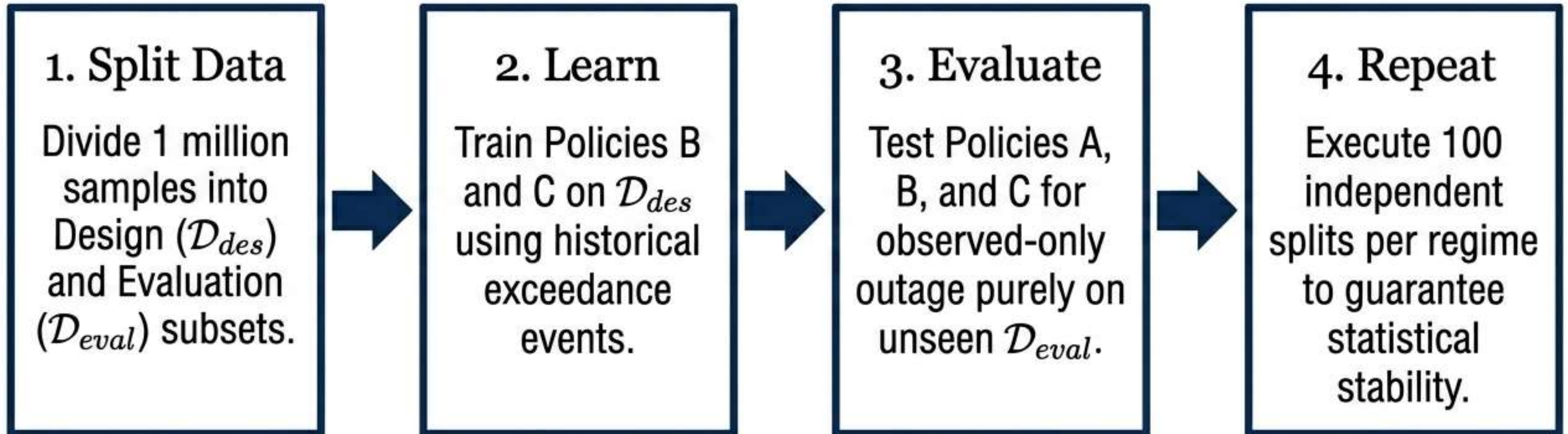
**Focus:** Complementarity and set-level diversity.



# Rigorous Monte Carlo Evaluation Protocol

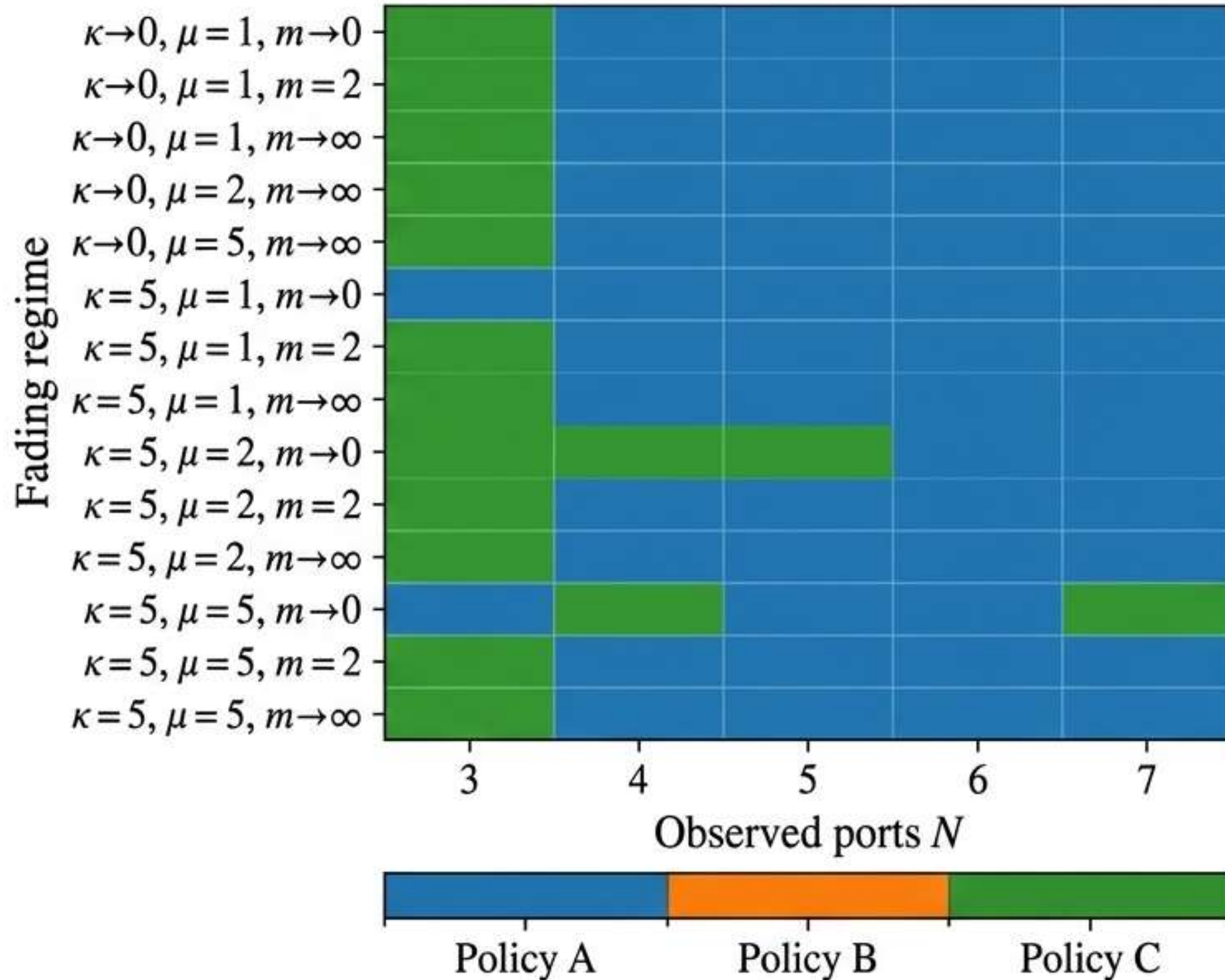
|                              |                    |                                      |                                       |
|------------------------------|--------------------|--------------------------------------|---------------------------------------|
| $S = 10^6$<br>samples/regime | $M = 100$<br>ports | Budgets<br>$N \in \{3, 4, 5, 6, 7\}$ | Target Threshold<br>$\eta_{th} = 0.8$ |
|------------------------------|--------------------|--------------------------------------|---------------------------------------|

## Algorithm 1: Monte Carlo Evaluation





# The Budget-Dependent Crossover Effect



## Tight Budget ( $N = 3$ )

Policy C (Greedy Coverage) wins in 12 of 14 regimes (85.7%). When measurement starved, explicitly searching for complementarity is optimal.

## Relaxed Budget ( $N \geq 4$ )

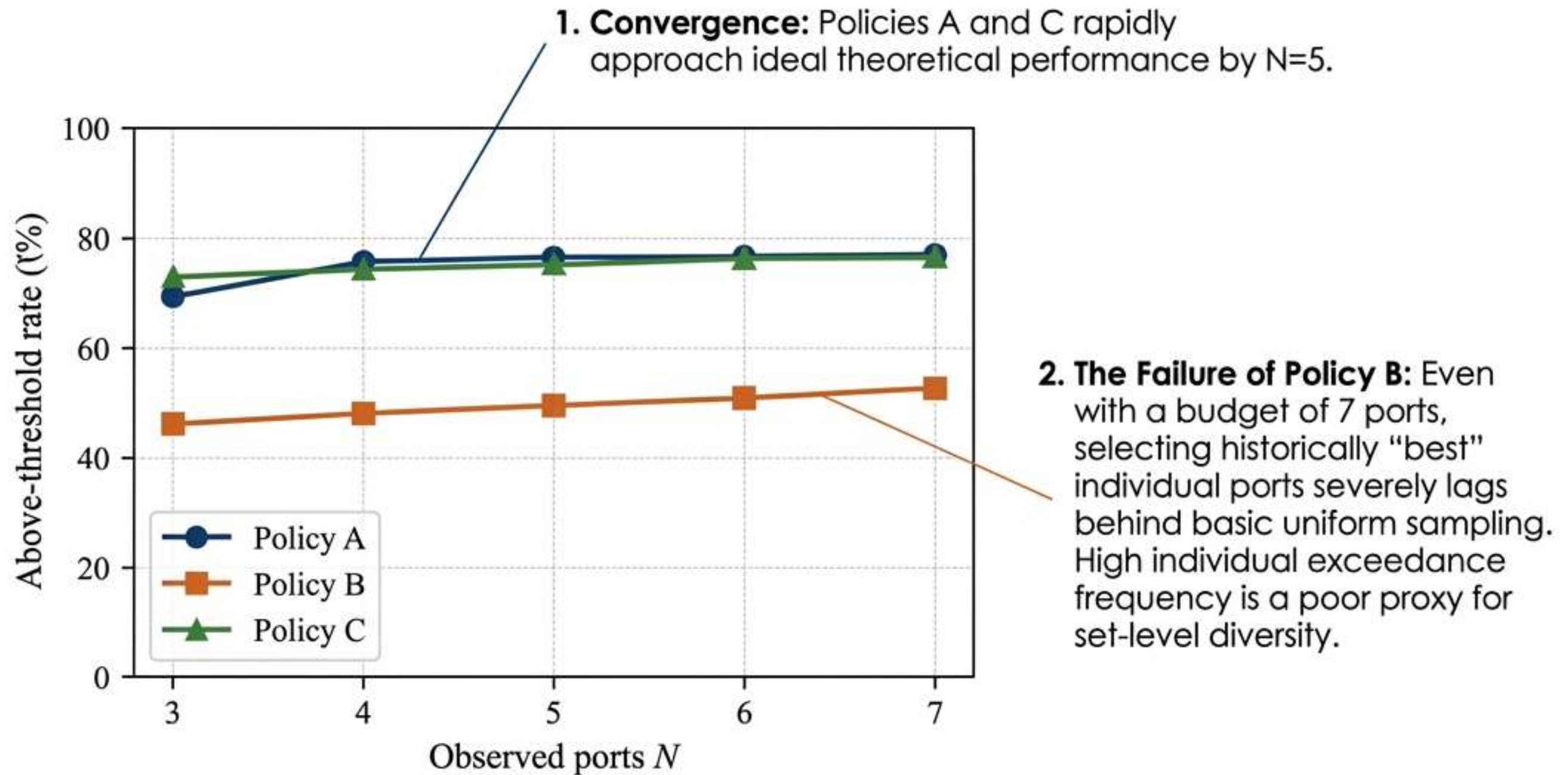
Policy A (Uniform Subsampling) completely dominates, hitting a 100% win rate at  $N = 6$ . A single additional observation fundamentally shifts the optimal strategy.

## The Anomaly

Policy B (Marginal Ranking) achieves a 0% win rate across all regimes and budgets.



# System Reliability vs. Observation Budget



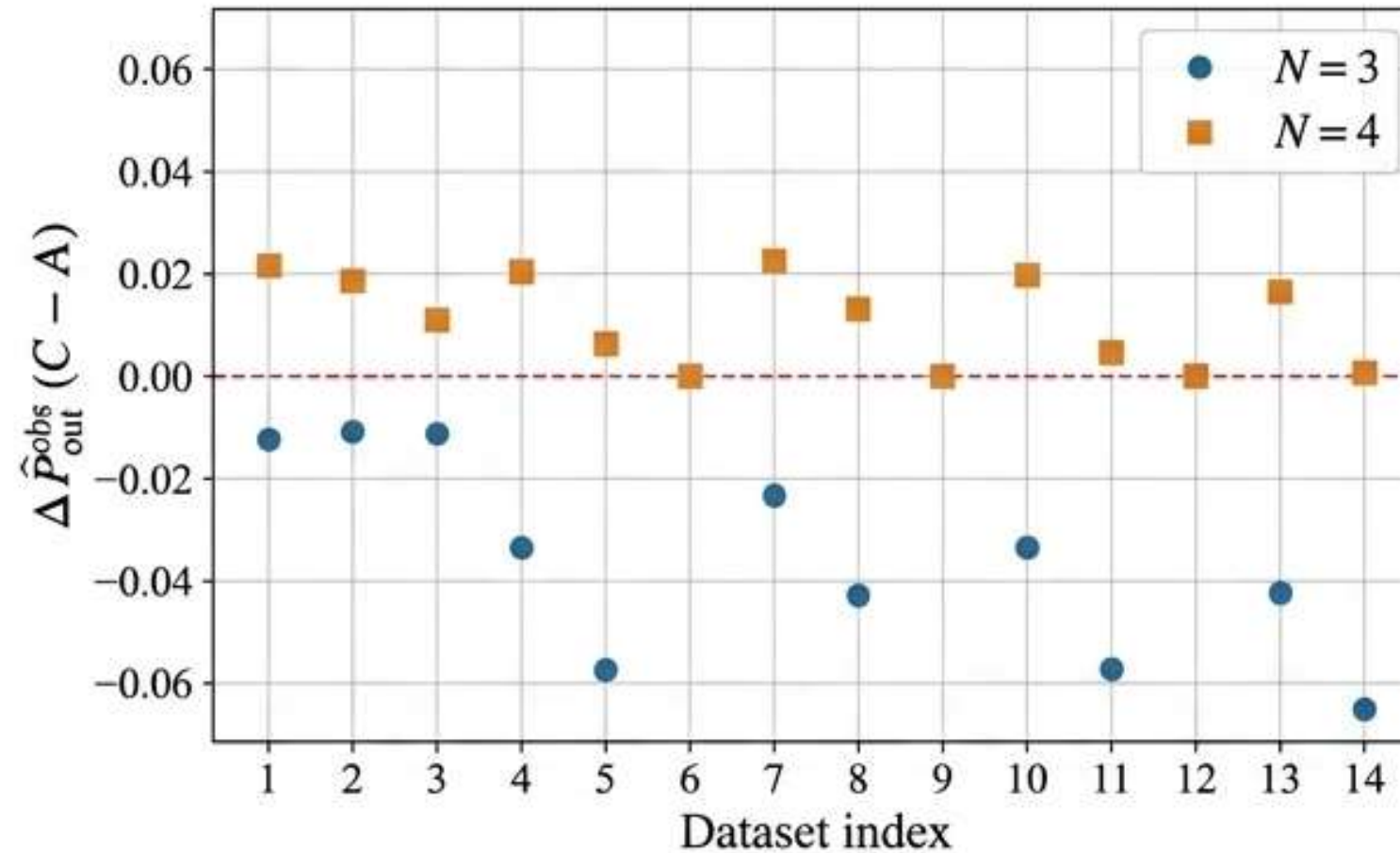


# Statistical Certainty of the N=3 to N=4 Crossover

## Blue Dots ( $N = 3$ ):

Consistently negative across datasets.

Greedy coverage is statistically superior under extreme starvation.



## Orange Squares ( $N = 4$ ):

Consistently positive across datasets.

The addition of a single observation fundamentally shifts the optimal strategy to **Uniform** coverage.

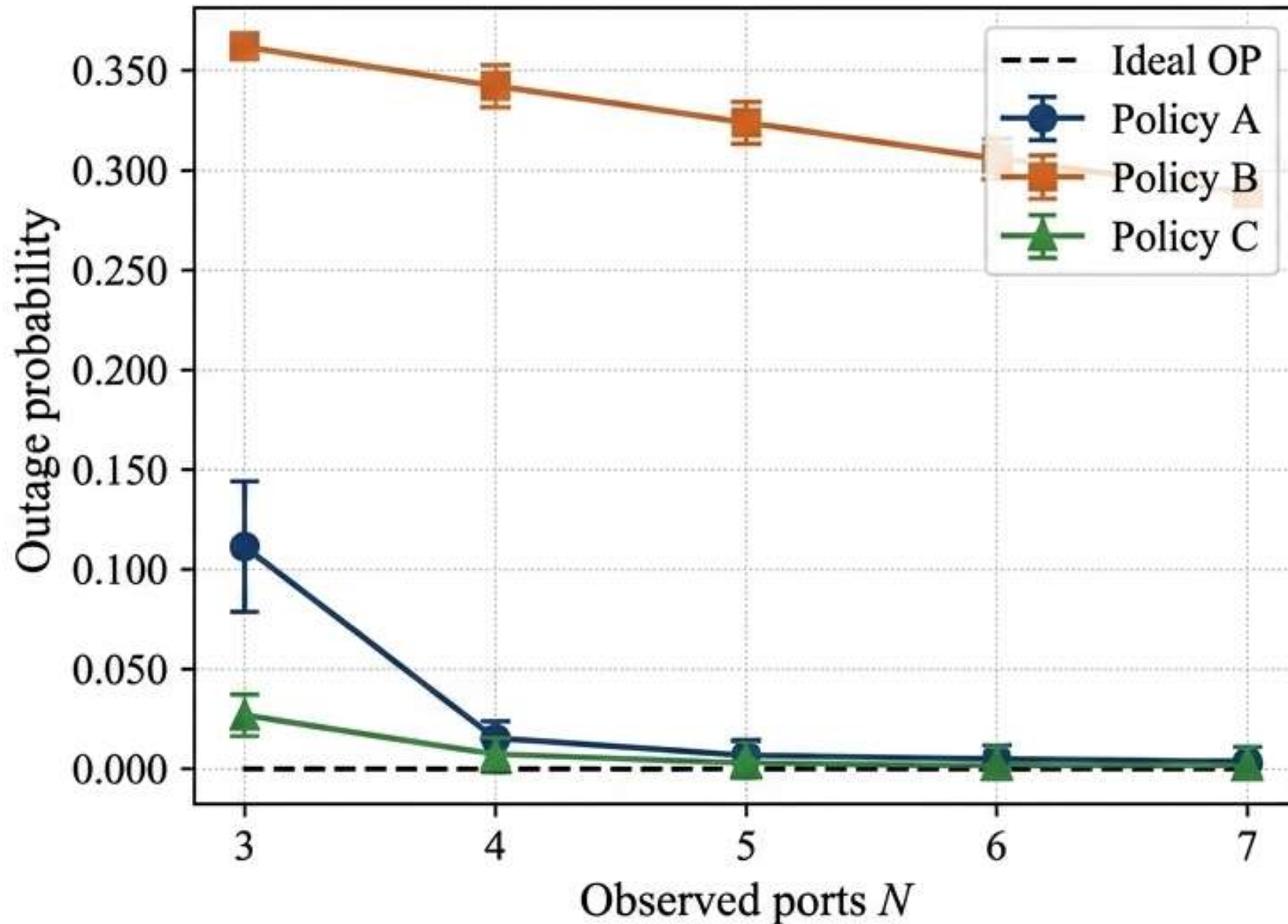
Metric Definition:  $\Delta \hat{P}_{out}^{obs} = C - A$

Negative values favor **Policy C**.  
Positive values favor **Policy A**.



# Approaching Ideal Performance Limits

(Scattered-field dominated regime:  $\kappa \rightarrow 0, \mu = 5, m \rightarrow \infty$ )



## Theoretical Floor

The black dashed line represents the Ideal Outage Probability—the absolute minimum possible outage if all 100 ports were measured.

## High Efficiency

Policies A and C drive the outage probability down to near-ideal levels with just 4 to 5 measurements.

## Absolute Stability

The 95% confidence intervals are smaller than the data markers, verifying the stability of the reading policies across 100 independent splits.



# The Computational Reality: Selection-Time Benchmark

Policy A (Uniform)

**0.0638 ms**

Mean Selection Time. Requires  
zero data-driven tuning.  
Instantaneous operation.

Policy C (Greedy)

**1,595.63 ms**

Mean Selection Time. Requires  
iterating through a massive  
design dataset.

**THE VERDICT:** Policy C is approximately 25,000x slower. For any budget  $N \geq 4$ , Policy A delivers optimal outage performance with practically instantaneous selection.

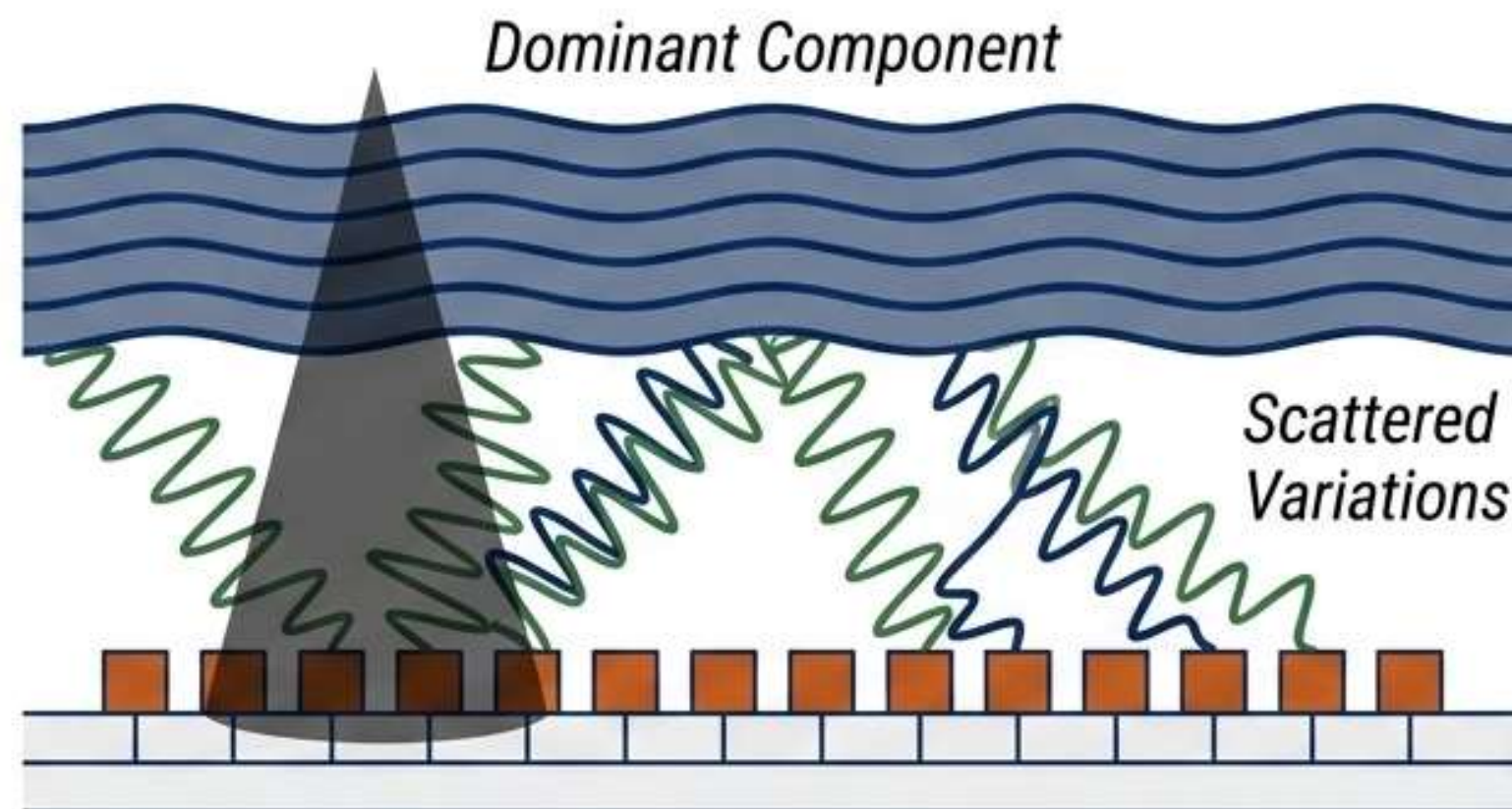


# The Physics of the Aperture: Why Marginal Ranking Fails

## The Mechanism

The dominant signal component shifts the entire aperture coherently. It does not create spatial diversity.

Spatial diversity is generated entirely by sub-wavelength scattered field variations.



## The Synthesis

Because all ports share similar marginal statistics, ranking them in isolation (Policy B) simply selects a clustered group of physically proximate ports. If a shadow drops on that cluster, all selected ports fail together.

A system must either blindly sample across the entire spatial variation (Policy A) or explicitly compute complementary variations (Policy C).



# Establishing Port Reading as a FAS Design Layer

## Recommendation 1: Extreme Constraints ( $N \leq 3$ )

**Use Greedy Coverage.** When strictly limited by hardware or latency, paying the computational cost explicitly secures complementary ports and minimizes observed outage.

## Recommendation 2: Operational Default ( $N \geq 4$ )

**Default to Uniform Reading.** It is highly robust, converges rapidly to ideal theoretical performance limits, and avoids massive computational training overhead.

## Recommendation 3: Avoid Marginal Selection

**Do not rank ports by historical individual quality.** Per-port individual success frequency does not equate to set-level spatial diversity due to severe sub-wavelength correlation.



# Future Directions & Acknowledgments

## Future Research Vectors



Varying Aperture Lengths ( $D = 0.5$ ,  $D = 2$ ) to test the boundary limits of spatial correlation.



Correlation-aware subset selection (e.g., maximizing covariance determinant) to achieve Policy C performance without training samples.



Joint sensing-policy optimization under strict hardware latency and switching constraints.

## Acknowledgments

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